

Agile Foundations & Definition of Terms

- **Epic:** A large component of work or project focus area that is too vast to complete in a single sprint. In this case study, epics represent core functional modules like *Data Pipeline Setup*, *Student Insights View*, and *Cafeteria Admin View*.
- **Story (User Story):** A small, discrete requirement written from the perspective of an end-user (e.g., a student like Rahul/Priya, or a cafeteria manager) that can be completed within a sprint cycle.
- **Sprint:** A fixed, time-boxed period during which a development team completes a specific set of committed tasks. This project employs **three 4-day sprints**, totaling a **12-day project duration**.
- **Story Point:** A relative metric representing the estimated effort, complexity, and risk required to complete a story, typically scored using a Fibonacci sequence (1, 2, 3, 5,).
 - **1** = Very Easy / Minimal effort
 - **2** = Normal / Low complexity
 - **3** = Moderate complexity
 - **5** = Difficult / High effort

1. Product Backlog & Velocity Architecture

Sprint 1: Foundation & Infrastructure

- **Epic 1: Data Pipeline Setup & Clean**
 - *USN-1:* Clean raw student survey dataset to fix missing breakfast and stress parameters. **(Story Points: 3, Priority: High)**
 - *USN-2:* Merge/Join transaction logs with survey records via common key variables. **(Story Points: 2, Priority: High)**
- **Epic 2: Metric Frameworks**
 - *USN-3:* Construct analytical calculated fields in Tableau for Focus and Stress indexes. **(Story Points: 2, Priority: Medium)**
- **Sprint 1 Total Effort:** $3 + 2 + 2 = 7$ Story Points

Sprint 2: Student Portal Dashboards

- **Epic 3: Student Insights View**
 - *USN-4:* Design an interactive dashboard linking skipped breakfast routines with drops in student concentration levels (Scenario 1: Rahul). **(Story Points: 5, Priority: High)**
 - *USN-5:* Create trend visualization layouts highlighting emotional stress indicators vs. comfort food reliance (Scenario 2: Priya). **(Story Points: 3, Priority: High)**
- **Sprint 2 Total Effort:** $5 + 3 = 8$ Story Points

Sprint 3: Administrative Insights & Deployment

- **Epic 4: Cafeteria Admin View**

- *USN-6*: Build high-level canvas comparing low fruit/vegetable intake with high fast-food volumes (Scenario 3: Cafeteria Manager). **(Story Points: 5, Priority: High)**
- *USN-7*: Introduce multi-dimensional dashboard filters covering academic majors and residential status. **(Story Points: 2, Priority: Medium)**
- **Sprint 3 Total Effort: 5 + 2 = 7 Story Points**

Velocity Profile

- **Total Project Story Points:** 7 + 8 + 7 = 22 Story Points
- **Total Sprints Completed:** 3 Sprints
- **Average Velocity per Sprint:**

$$\text{Sprint Velocity} = \frac{22 \text{ Story Points}}{3 \text{ Sprints}} = 7.33 \text{ Story Points per Sprint}$$

2. Burn-Down Tracking Metrics

The operational pacing of the project requires calculation of the Average Velocity (AV) per calendar unit. This dictates how many story points must cross the definition of "Done" daily to satisfy timelines.

Daily Burn Formula

$$\text{AV per Day} = \frac{\text{Total Story Points}}{\text{Total Project Duration in Days}} = \frac{22 \text{ points}}{12 \text{ Days}} = 1.83 \text{ s}$$

Burndown Milestones & Actual Progress Tracking

This tracking grid establishes the work remaining timeline plotted across your active schedule:

- **Day 0 (Kickoff): 22 story points remaining** as the base backlog backlog is loaded.
- **Day 4 (End of Sprint 1): 15 story points remaining.** Infrastructure finalized, raw tables cleaned, and primary index fields configured in Tableau (USN-1, USN-2, USN-3 complete).
- **Day 8 (End of Sprint 2): 7 story points remaining.** Student behavior tracking frameworks and interactive stress timelines deployed (USN-4, USN-5 complete).
- **Day 12 (End of Sprint 3): 0 story points remaining.** Cafeteria administration panels, menu optimization filters, and deployment are fully complete (USN-6, USN-7 complete).